Extra Practice 1: Factoring Review: Common Factors and $x^2 + bx + c$

A. Skills

1. Factor.
(a) $18x^4 - 27x^3$ (b) $6a^3b - 9a^2b^2 + 12ab^3$ (c) $-10y^3 - 25y^2$

(d) $4x(x - 7) - 9(x - 7)$ (e) $2y(3y + 1) + (3y + 1)$ (f) $m(4 - m) + 6(m - 4)$

2. Factor by grouping.
(a) $x^3 - 3x^2 + 4x - 12$ (b) $10xy + 5x - 6y - 3$ (c) $ab - 3a + 7b - 21$

3. Factor where possible.
(a) $x^2 + 11x + 18$ (b) $x^2 - 12x + 32$ (c) $x^2 + 5x - 36$

(d) $x^2 - 3x - 54$ (e) $a^2 - 15a + 56$ (f) $x^2 + 6x + 7$

4. Factor.
(a) $x^2 + 10xy + 24y^2$ (b) $a^2 - 6ab - 27b^2$ (c) $m^2 - 14mn + 45n^2$

B. Applying

5. Factor fully.
(a) $6x^2 - 6x - 72$ (b) $3a^3 + 24a^2 + 45a$ (c) $-y^2 + 9y - 20$

(d) $24 - 5x - x^2$ (e) $2x^2y - 16xy + 30y$ (f) $x^4 - 2x^3 - 35x^2$

6. Which of these polynomials have $x - 2$ as a factor? Show your factoring.

(a) $x^2 + 3x - 10$

(b) $4x^2 - 4x - 8$

(c) $x^2 - 4x - 12$

(d) $x^3 - 2x^2 + 7x - 14$

7. A rectangle has an area of $(x^2 + 12x + 32)$ cm^2 and a width of $(x + 4)$ cm.
(a) Find an expression for its length.

(b) The perimeter is 64 cm. Find x and the rectangle's dimensions.

8. The fully factored form of $3x^2 - 3x - 60$ is
A. $(3x - 15)(x + 4)$ **B.** $3(x - 5)(x + 4)$ **C.** $3(x + 5)(x - 4)$ **D.** $(x - 5)(3x + 12)$

9. Numerical Response The trinomial $x^2 + 8x + c$, where c is a positive integer, can be factored over the integers for several values of c. The sum of all these values of c is _____.

C. Stretch

10. Find the value of k so that $x - 4$ is a factor of $x^2 + kx + 36$.

11. Factor fully: $3a(a + 2) - 5(a + 2)^2$.

12. Factor: $12 - 4y - 3x + xy$.

13. Explain why $n^2 + 5n + 6$ is even for every integer n.

Extra Practice 2: Factoring $ax^2 + bx + c$

A. Skills

1. Factor.
(a) $2x^2 + 13x + 15$ (b) $3x^2 - 11x + 10$ (c) $4x^2 + 4x - 3$

(d) $6x^2 - 5x - 6$ (e) $5x^2 - 21x + 4$ (f) $8x^2 + 10x - 3$

2. Factor fully.
(a) $4x^2 + 22x + 10$ (b) $9y^3 - 21y^2 - 18y$ (c) $30a^2 - 5a - 10$

3. Factor.
(a) $2x^2 + 9xy + 10y^2$ (b) $3a^2 - 10ab - 8b^2$ (c) $6m^2 - 17mn + 12n^2$

B. Applying

4. Factor fully.
(a) $-2x^2 - 5x + 12$ (b) $10 + 3x - 4x^2$

5. Factor fully.
(a) $\frac{1}{3}x^2 + 2x - 9$ (b) $\frac{1}{4}x^2 - x - 3$ (c) $\frac{2}{5}x^2 + \frac{6}{5}x + \frac{4}{5}$

6. (a) Is $9x^2 - 42x + 49$ a perfect-square trinomial? If so, factor it.

(b) Fill in the blank to make a perfect square: $36x^2 + _____ + 25$.

(c) Factor fully: $50x^2 + 40x + 8$.

7. A rectangle has an area of $(8x^2 + 2x - 15)$ m^2.

(a) Factor to find expressions for its dimensions.

(b) Write and simplify an expression for its perimeter.

(c) Find the dimensions and perimeter when $x = 5$.

8. Which binomial is a factor of $6x^2 + 7x - 3$?

A. $3x + 1$ **B.** $2x - 3$ **C.** $3x - 1$ **D.** $6x - 1$

9. Numerical Response The trinomial $5x^2 - 7x - 6$ can be written as $(5x + a)(x - b)$, where a and b are positive integers. Record a in the first box and b in the second box.

C. Stretch

10. Find all integers k so that $3x^2 + kx + 4$ can be factored over the integers.

11. A rectangle has an area of $(6x^2 + 5x - 6)$ cm^2 and a length of $(3x - 2)$ cm. Find its width. For what value of x is the rectangle a square? What is the square's area?

12. Factor fully: $0.4x^2 - 1.4x - 0.8$.

13. Find all positive integers k so that $kx^2 + 7x + 3$ can be factored over the integers.

Extra Practice 3: Factoring by Substitution: $a(f(x))^2 + b(f(x)) + c$

A. Skills

1. Factor fully.

(a) $x^4 + 7x^2 + 10$

(b) $b^4 - 4b^2 - 45$

(c) $x^6 - x^3 - 20$

(d) $a^2b^2 - 11ab + 28$

(e) $x^4 - 26x^2 + 25$

(f) $2y^4 + 10y^2 - 48$

2. Factor fully.

(a) $2x^4 - 7x^2 + 3$

(b) $9x^4 - 10x^2 + 1$

(c) $3p^6 - 11p^3 - 4$

(d) $10x^2y^2 + 9xy + 2$

(e) $3x^5 + 6x^3 - 45x$

3. Factor fully. Simplify each factor.

(a) $(x-1)^2 + 6(x-1) + 8$

(b) $(2a+1)^2 - (2a+1) - 12$

(c) $2(x+4)^2 - 5(x+4) - 3$

(d) $9(x-2)^2 + 12(x-2) + 4$

B. Applying

4. Factor $2\sin^2 x - 5\sin x - 3$.

5. Factor fully: $3(x+2)^2 + 3(x+2) - 18$.

6. Factor fully: $5(1-2x)^2 + 7(1-2x) - 6$.

7. Factor $3(x+1)^2 - 10(x+1) + 8$ by substitution, then check by expanding first.

8. Which expression is a factor of $2x^4 + x^2 - 6$?

A. $2x^2 + 3$ **B.** $x^2 + 2$ **C.** $x^2 - 2$ **D.** $2x - 3$

9. Numerical Response When factored, $(x-5)^2 + 4(x-5) - 12 = (x+a)(x-b)$, where $a > 0$ and $b > 0$. The value of $a + b$ is _____.

C. Stretch

10. Factor fully: $x^4 - 10x^2y^2 + 9y^4$.

11. Factor fully: $3(x+1)^2 - 2(x+1)(x-1) - (x-1)^2$.

12. Find all integers b for which $2x^4 + bx^2 - 5$ can be factored as $(2x^2 + p)(x^2 + q)$, with p and q integers.

Extra Practice 4: Differences of Squares, Including $a^2(f(x))^2 - b^2(g(y))^2$

A. Skills

1. Factor fully, if possible.

(a) $4x^2 - 121$

(b) $100a^2 - 49b^2$

(c) $16x^2y^2 - 1$

(d) $3x^2 - 75$

(e) $45m^2 - 20n^2$

(f) $x^2 + 64y^2$

(g) $\frac{1}{25}x^2 - \frac{1}{4}$

(h) $0.81 - y^2$

(i) $-8a^2 + 2$

2. Factor fully.

(a) $y^4 - 16$

(b) $625a^4 - b^4$

(c) $5x^4 - 5$

(d) $x^4y^2 - 9$

3. Factor fully. Simplify each factor.

(a) $(x + 6)^2 - 16$

(b) $25 - (2y - 1)^2$

(c) $(x + 5)^2 - (x - 3)^2$

(d) $4(x + 1)^2 - 9(y - 1)^2$

B. Applying

4. A rectangular poster has an area of $25x^2 - 16y^2$ square centimetres; its length and width are of the form $ax + by$ with

integers a and b. The perimeter is 60 cm and the length is 1.5 times the width. Find x, y and the dimensions.

5. A square picture frame has outer side length $(3x + 2)$ cm and a square opening of side length $(x + 4)$ cm.
(a) Write and fully factor an expression for the area of the frame.

(b) Find the area when $x = 5$.

6. Factor fully.
(a) $3x^3 - 12x^2 - 15x$ (b) $x^3 - 5x^2 - 4x + 20$

(c) $4x^4 - 13x^2 + 9$ (d) $16(x - 1)^2 - 25$

7. Which binomial is a factor of both $2x^2 + 5x - 12$ and $4x^2 - 9$?
A. $2x + 3$ **B.** $2x - 3$ **C.** $x + 4$ **D.** $x - 4$

8. Numerical Response The expression $49(x + 1)^2 - 9(x - 3)^2$ can be factored as $k(x + a)(bx - c)$, where k, a, b and c are natural numbers. The value of $k + a + b + c$ is _____.

C. Stretch

9. Factor fully: $x^2 - 2xy + y^2 - 9z^2$.

10. Factor fully: $a^4 - (3a - 2)^2$.

11. Use a difference of squares to evaluate without a calculator. (a) $103^2 - 97^2$ (b) $2026^2 - 2024^2$

Extra Practice 5: Solving Quadratic Equations by Factoring

A. Skills

1. Solve.

(a) $(x+9)(2x-1)=0$

(b) $x^2-13x=0$

(c) $100x^2-9=0$

(d) $x^2-10x+21=0$

(e) $x^2+x-56=0$

(f) $3x^2+14x-5=0$

(g) $4x^2+28x+49=0$

(h) $-x^2+6x+16=0$

(i) $6x^2+5x-6=0$

2. Write each equation in standard form, then solve.

(a) $5x^2=2-9x$

(b) $x(2x+1)=15$

(c) $(x+3)^2=4x+17$

(d) $\frac{1}{2}x^2+2x=16$

3. Write a quadratic equation in standard form, with integer coefficients, that has the given roots. (a) -4 and 9 (b) $\frac{5}{3}$ and $-\frac{1}{2}$ (c) a double root of 4

B. Applying

4. Determine the zeros of $y=3x^2+3x-18$ and state the x-intercepts of its graph.

5. The product of two consecutive integers is 41 more than their sum. Determine the integers.

6. A rectangular lawn 15 m by 10 m is surrounded by a path of uniform width. The area of the path alone is 84 m^2.

How wide is the path?

7. A model rocket is launched from the ground. Its height after t seconds is $h = -5t^2 + 40t$ metres.
(a) At what times is the rocket 75 m high?

(b) When does it land?

8. The length of a rectangle is 7 cm more than its width, and its diagonal is 13 cm. Determine the dimensions. (Use the Pythagorean theorem.)

9. Which equation has a double root?
A. $x^2 - 9 = 0$ **B.** $x^2 + 6x + 9 = 0$ **C.** $x^2 - 6x = 0$ **D.** $x^2 + 9x = 0$

10. Numerical Response The arithmetic series $5 + 9 + 13 + \ldots$ has a sum of 230. The number of terms is _____.

C. Stretch

11. One root of $2x^2 + kx - 15 = 0$ is $x = 3$. Determine k and the other root.

12. Solve $x^4 - 10x^2 + 9 = 0$. (Factor as a trinomial in x^2 first.)

13. Squares of side 3 cm are cut from the corners of a square sheet of cardboard, and the sides are folded up to make an open box with a volume of 300 cm^3. Determine the side length of the original sheet.

Extra Practice 6: Solving Radical Equations and Checking for Extraneous Roots

A. Skills

1. State the restriction on the variable.
(a) $\sqrt{x-11}$ (b) $\sqrt{9-4x}$ (c) $\sqrt{2x+7}+\sqrt{x}$ (d) $\sqrt{x+1}=\sqrt{6-3x}$

2. Solve and verify. State "no solution" where it applies.
(a) $\sqrt{x+8}=5$ (b) $\sqrt{6x-2}-2=2$ (c) $5\sqrt{x-1}=15$

(d) $8-2\sqrt{x+3}=0$ (e) $\sqrt{4x+5}+7=3$ (f) $\sqrt{\dfrac{2x+1}{5}}=3$

3. Solve and verify.
(a) $\sqrt{6x-5}=\sqrt{2x+11}$ (b) $\sqrt{1-2x}=\sqrt{x-8}$ (c) $3\sqrt{x}=\sqrt{x+16}$

B. Applying

4. The time for an object to fall d metres is $t=\sqrt{\dfrac{d}{4.9}}$ seconds. How far does an object fall in 3 s?

5. A square has side length $\sqrt{3x+4}$ cm and perimeter 20 cm. Determine x.

6. Using $T=2\pi\sqrt{\dfrac{L}{9.8}}$, determine the length of a pendulum with a period of 1.5 s, to the nearest hundredth of a metre.

7. Explain why $\sqrt{3x-1}+\sqrt{x+4}=-2$ has no solution.

8. The solution of $\sqrt{2x-3} = \sqrt{x+1}$ is

A. $x = 4$ **B.** $x = -4$ **C.** $x = \frac{2}{3}$ **D.** no solution

9. Numerical Response The root of $2\sqrt{5x-1} - 3 = 5$ is $x =$ _____.

C. Stretch

10. The equation $\sqrt{ax-2} = 4$ has the root $x = 6$. Determine a.

11. For which values of c does $\sqrt{x-3} + c = 1$ have no solution? Explain.

12. (Extension: cube roots.) Solve $\sqrt[3]{2x-3} = -3$. Explain why a cube-root equation has no restriction and why cubing does not create extraneous roots.

Extra Practice 7: Radical Equations That Lead to Quadratics

A. Skills

1. State the restrictions on x.
(a) $\sqrt{x+2} = x-4$ (b) $\sqrt{6-2x} = x$ (c) $\sqrt{x+3} + x = 3$

2. Solve and verify. Identify any extraneous roots.
(a) $\sqrt{x+30} = x$ (b) $\sqrt{7x-6} = x$

(c) $\sqrt{3x+7} = x-1$ (d) $\sqrt{11-x} = x+1$

3. Isolate the radical, then solve and verify.
(a) $x - \sqrt{x+10} = 2$ (b) $2\sqrt{x+3} - x = 0$ (c) $\sqrt{4x+1} - 1 = x$

B. Applying

4. Solve $x - 6\sqrt{x} + 8 = 0$ by letting $u = \sqrt{x}$. Verify both roots.

5. Solve and verify.

(a) $\sqrt{x+4} - \sqrt{x-4} = 2$

(b) $\sqrt{2x+6} - \sqrt{x+4} = 1$

6. A number increased by twice its square root is 24. Write a radical equation and determine the number.

7. A rectangular garden plot is x m wide and $(x + 14)$ m long. A straight path along its diagonal is $(x + 16)$ m long.

(a) Write a radical equation for the length of the diagonal.

(b) Solve it and state the dimensions of the plot. Explain why one root is rejected.

8. The extraneous root of $x = \sqrt{5x + 14}$ is

A. -7 **B.** -2 **C.** 2 **D.** 7

9. Numerical Response The root of $2x - \sqrt{6x - 5} = 3$, to the nearest tenth, is _____.

C. Stretch

10. Solve $\sqrt{x-3}+\sqrt{x}=\sqrt{2x+1}$ and verify.

11. Solve $\sqrt{3x+1}=\sqrt{x+4}+1$ and verify.

12. (a) Explain, without solving, why $\sqrt{2x+1}+3=0$ has no solution.

(b) Explain why $\sqrt{x-5}+\sqrt{2-x}=1$ has no solution.

(c) Find the value of c so that $\sqrt{x+c}=x$ has the root $x=5$. Then find the extraneous root that squaring produces.

Extra Practice 8: Unit 4 Review — Practice Test

Give every answer exact. Verify the roots of every radical equation.

A. Multiple Choice

1. When $3x^2 - 12$ is factored fully, the result is
A. $(3x - 6)(x + 2)$ **B.** $3(x - 2)(x + 2)$ **C.** $3(x - 2)^2$ **D.** $3(x - 4)(x + 4)$

2. One factor of $10x^2 + 13x - 3$ is
A. $2x - 3$ **B.** $5x + 1$ **C.** $5x - 1$ **D.** $10x - 1$

3. The trinomial $x^2 + kx + 18$ can be factored over the integers for each of these values of k **except**
A. 9 **B.** 11 **C.** -19 **D.** 7

4. When $2x^4 - 32$ is factored fully, the result is
A. $2(x^2 - 4)(x^2 + 4)$
B. $2(x - 2)(x + 2)(x^2 + 4)$
C. $2(x - 2)^2(x + 2)^2$
D. $(2x^2 - 8)(x^2 + 4)$

5. When factored, $(x - 2)^2 - 5(x - 2) - 14$ is
A. $(x - 7)(x + 2)$ **B.** $x(x - 9)$ **C.** $(x - 9)(x + 4)$ **D.** $(x - 5)(x + 4)$

6. A quadratic equation with roots $-\frac{3}{2}$ and 4 is
A. $2x^2 + 5x - 12 = 0$
B. $2x^2 - 5x - 12 = 0$
C. $2x^2 - 11x + 12 = 0$
D. $2x^2 - 5x + 12 = 0$

7. The x-intercepts of the graph of $y = x^2 + 2x - 15$ are
A. $(5, 0), (-3, 0)$ **B.** $(-5, 0), (3, 0)$ **C.** $(0, -15)$ **D.** $(-15, 0), (1, 0)$

8. All the restrictions on x in the equation $\sqrt{2x + 9} = x - 3$ are
A. $x \geq -\frac{9}{2}$ **B.** $x \geq 3$ **C.** $-\frac{9}{2} \leq x \leq 3$ **D.** $x \geq 0$

9. The solution of $\sqrt{x + 1} = x - 5$ is
A. $x = 3$ only **B.** $x = 8$ only **C.** $x = 3, 8$ **D.** no solution

10. The equation $\sqrt{x+3}+5=2$ has no solution because

A. the radicand can be negative

B. after isolating, a square root would have to equal -3

C. squaring both sides gives a quadratic

D. 5 is greater than 2, so x must be negative

B. Numerical Response

11. Numerical Response The expression $8x^2-2x-15$ can be written as $(ax-b)(cx+d)$, where a, b, c and d are positive integers. Write a, b, c and d in the four boxes, in that order.

12. Numerical Response The positive root of $6x^2-x-12=0$, to the nearest tenth, is _____.

13. Numerical Response A 12 m by 8 m pool is surrounded by a deck of uniform width. The area of the deck alone is 96 m^2. The width of the deck, in metres, is _____.

14. Numerical Response The root of $\sqrt{2x-1}+2=x$ is _____.

C. Written Response

15. Factor fully.

(a) $48x^2-3y^2$

(b) $-2x^3+4x^2+30x$

(c) $4(x+2)^2-(x-1)^2$

(d) $2x^3-3x^2-8x+12$

16. Solve by factoring.

(a) $4x^2 + 9 = 12x$

(b) $x(2x + 1) = 3(x + 4)$

17. A ball is kicked from the top of a hill. Its height above the bottom of the hill, h metres, after t seconds is $h = -5t^2 + 15t + 20$.

(a) When does the ball reach the bottom of the hill ($h = 0$)?

(b) At what times is the ball 30 m high? Explain why there are two times.

18. Consider the radical equation $\sqrt{x} - \sqrt{2x - 2} = 1$.

(a) Determine the restrictions on x.

(b) Explain why an extraneous root may appear when this equation is solved algebraically.

(c) Algebraically determine the root(s) of the equation.